

Sami Hatoum

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Education

University College London | MSc Artificial Intelligence and Imaging 2025 – 2026

Predicted Distinction

Generalisation Gap Decomposition for Image Segmentation (MSc Dissertation)

Modules: Applied Deep Learning, Machine Learning in Medical Imaging, Statistical Modelling, Inverse Problems

University of St Andrews | BSc Computer Science

2021 – 2025

First Class Honours, Dean's List, Honours Project Distinction

Dissertation: *Sokoban Procedural Instance Generation* (constraint optimisation over a PSPACE-complete design space)

Modules: Data Structures and Algorithms, Databases, Computational Complexity, Constraint Programming

Skills

Languages: Python, Java, JavaScript, TypeScript, C++, SQL

Web: React, Next.js, Node.js, Flask

Cloud & DevOps: AWS (Certified Cloud Practitioner), Docker, Git, GitHub Actions, Linux

Databases & Testing: PostgreSQL, Redis, Jest, Pytest, Locust, Supertest

Machine Learning: PyTorch, scikit-learn, Hugging Face Transformers, MONAI

Experience

PricewaterhouseCoopers (PwC)

Software Engineering Intern

Aug 2024 – Sep 2024

- Shipped an internal **client onboarding dashboard** in **React** and **Node.js** replacing a manual spreadsheet workflow for a financial services team; pilot timings showed onboarding falling from **3 days to 4 hours** across 200+ onboarded users
- Designed a **PostgreSQL** schema with **role-based access control** across 4 user roles, working with a data scientist on requirements; integrated with PwC's internal **SSO** via **OAuth 2.0**
- Wrote **CI/CD pipelines** in **GitHub Actions** with linting, automated test gates, and staged deployments to **AWS**; no failed releases across the deployment window post-adoption
- Achieved **90%+ code coverage** across **45 unit and integration tests** spanning the REST API and data access layers

Software Engineering Intern

Jun 2023 – Jul 2023

- Owned the data pipeline and evaluation harness for a **PyTorch BERT** sentiment classifier processing **10,000 daily customer reviews** for a retail client; reached **F1 0.91** on the held-out test split
- Deployed as a containerised **Flask REST API** on **AWS EC2** with gunicorn workers behind an Application Load Balancer; sustained **150 req/s** with **p95 latency under 500ms** in Locust load tests
- Built **CloudWatch** dashboards on prediction drift and request volume, defining alert thresholds as rollback triggers
- Optimised preprocessing with **pandas** vectorisation, cutting daily batch processing time by **60%**

Projects

Real-Time Collaborative Code Editor

- Built a multi-user code editor in **React** and **Node.js** using **WebSockets** and **Operational Transform** for conflict-free concurrent editing, with syntax highlighting for 12 languages via CodeMirror
- Implemented a **Redis** pub/sub layer for horizontal scaling across server instances, sustaining **50+ concurrent editors** at **p95 synchronisation latency under 100ms** on k6 benchmarks, with automatic reconnection on network loss
- Added **JWT authentication**, per-room permission levels (owner/editor/viewer), and session persistence in **PostgreSQL** enabling users to rejoin documents with full edit history intact

Database Query Optimiser

- Built a **PostgreSQL** analysis tool in **Python** parsing query ASTs and **EXPLAIN plan** output to surface missing indexes; Dockerised for reproducibility
- Across **200+ generated queries** over a 5M-row, 12-table schema, identified **21 missing indexes** cutting average query time by **65%** (worst case 2.4s to 420ms) and removing sequential scans on hot paths

Generalisation Gap Decomposition for Image Segmentation (MSc Dissertation)

- Decomposing the internal-to-external generalisation gap in pelvic MRI segmentation with **data-scaling learning curves**, separating finite-sample error from irreducible distribution shift on an independent multi-scanner cohort
- Built the supporting selective-prediction pipeline (**nnU-Net** with uncertainty-based deferral of low-confidence scans to clinical review), reaching **0.89 Dice at 85% coverage**

Leadership

St Andrews AI Society | Machine Learning Officer

Sep 2024 – Apr 2025

- Led weekly machine learning workshops for **50+ members** spanning technical and non-technical backgrounds, designing sessions tractable across a wide range of mathematical and programming experience